Do Now · retrieval

A*: heuristic reliability and route evaluation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Calculate f when g=4 and h=3.

Q2

What is the role of a predecessor table?

Q3

If S–A=2 and A–G=5, what is this route's total cost?

Q4

The true cheapest remaining cost from A is 4. Is h(A)=6 admissible?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find the real remaining cost — From A, the cheapest route A→B→G costs 2+2=4. Therefore h(A)=3 is admissible, but h(A)=9 is not.

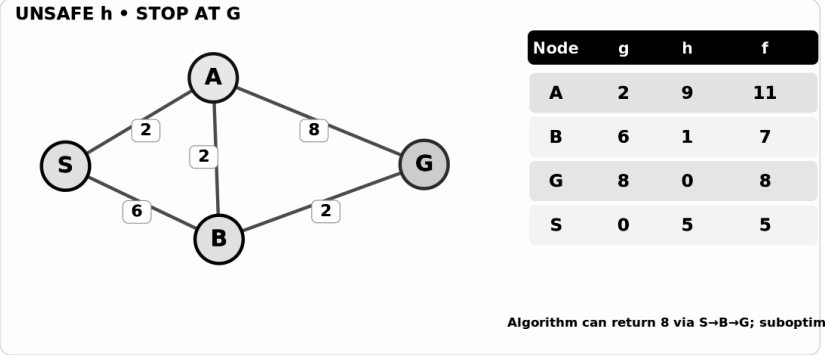
Core idea 2

Safe estimates support optimal search — With h(A)=3, A is expanded before B, improving the route to B and finally G at cost 6.

Core idea 3

Watch the misleading estimate — With h(A)=9, A has f=11; B has f=7 and leads to goal at cost 8.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Why the error matters — A* that stops on the first goal removed from the open set can return this 8-cost path before exploring A, even though a 6-cost route exists.

Core idea 5

Check the guarantee — Admissibility means no overestimate. Consistency adds the edge test h(u) ≤ weight(u,v) + h(v). Without reopening settled nodes, A* requires consistency for its shortest-path guarantee. Explain this as well as the route cost.

Vocabulary · use the correct technical terms

admissible — Heuristic never overestimates the true cheapest remaining cost.

consistent — Heuristic obeys h(u) ≤ edge(u,v)+h(v) for every edge.

suboptimal — A valid route whose cost is greater than the cheapest possible.

frontier — Discovered vertices awaiting expansion.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Check admissibility and trace U→G with safe estimates.

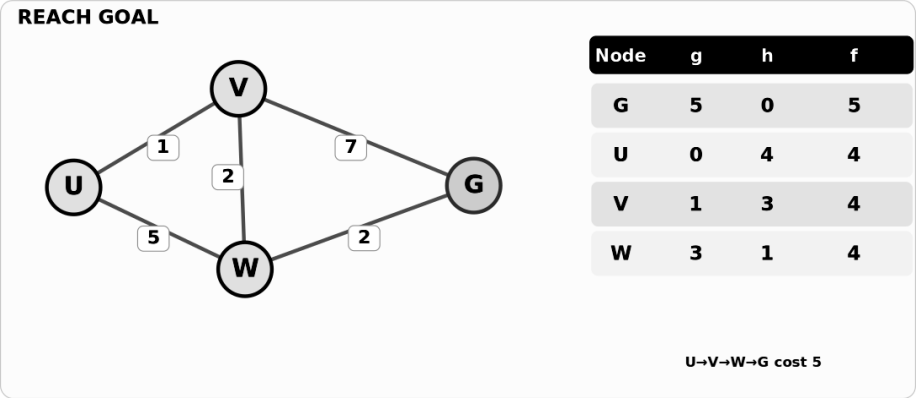
Follow the worked method

1. Predict the choice — V has g1,h3,f4; W has g5,h1,f6. Expand V.

2. Review the choice — W improves the goal G to g5,h0,f5.

3. Review the choice — The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

Source model · final illustrated step



Node	g	h	f
G	5	0	5
U	0	4	4
V	1	3	4
W	3	1	4

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the choice
After M, O has f5+1=6; N has f2+7=9, so O is selected.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

The true remaining distance from node A to goal G is 6. Decide whether h(A)=4 and h(A)=8 are admissible, with reasons.

Question 2

Edges S–A=2, S–B=6, A–B=2, B–G=2, A–G=8. Calculate the best S→G route and explain why h(A)=9 can mislead A*.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare Dijkstra's algorithm and A* in their selection rule, heuristic use and guarantee of shortest paths.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If a heuristic overestimates the true remaining cost, A* may lose its shortest-route guarantee.

Use key vocabulary from memory

1. Explain admissible in your own words.

2. Explain consistent in your own words.

3. Explain suboptimal in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.